



The first three *Church numerals*, and two terms mimicking addition and multiplication on them, are

$$\begin{aligned} \text{zero} &:= \lambda f x. x, & \text{one} &:= \lambda f x. f x, & \text{two} &:= \lambda f x. f (f x), \\ \text{add} &:= \lambda m n f x. m f (n f x), & \text{mult} &:= \lambda m n f x. m (n f) x. \end{aligned}$$

- (a) Show that $\text{add one one} \rightarrow_{\beta} \text{two}$.
 (b) Prove that $\text{add one one} \not\rightarrow_{\beta} \text{mult one zero}$.

Nederpelt & Geuvers, *Type Theory and Formal Proof* (CUP, 2014), Exercise 1.10.

I Definitions

Reduction (Defs. 1.8.1, 1.8.3, 1.8.5). One step is $(\lambda x. M) N \rightarrow_{\beta} M[x := N]$, applied anywhere inside a term; $M \rightarrow_{\beta} N$ is a chain of zero or more such steps, and $M =_{\beta} N$ is a chain in which each step may go either way.

Normal form (Def. 1.9.1). M is in β -normal form if it contains no redex $(\lambda x. P) Q$.

Church–Rosser (Thm. 1.9.8). If $M \rightarrow_{\beta} N$ and $M \rightarrow_{\beta} L$, then $N \rightarrow_{\beta} R$ and $L \rightarrow_{\beta} R$ for some R . Hence a term has at most one β -normal form (Lem. 1.9.10(2)).

Name _____

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Answer

Out of room? Continue on the back of the page.